

Personal Statement

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I learned the cost of an unbehavioural transport system the slow way. Growing up in Bandung, I watched a city of three million people negotiate two-lane streets designed for a third of that. Each administration proposed the same technical answer (more flyovers, more toll roads), and each time the data showed the new capacity absorbed within five years, a textbook example of induced travel demand that classical four-step traffic-engineering models routinely under-specify. When I joined Aston Business School for an MSc in Business Analytics under Dr. Ammar Al-Bazi, the Decision Models, Software Analytics, and Effective Management Consultancy modules formalised an intuition I had carried for a decade: that the most consequential decisions in a transport and logistics system are made by its people, not by its planners. This PhD is the logical extension of that intuition.

Academic foundations in operations research

My academic journey began with a Bachelor of Business Administration in Marketing at Ritsumeikan Asia Pacific University, Japan, where I held the JASSO Scholarship and the APU 50% Tuition Reduction Scholarship. APU laid my foundation in consumer behaviour, marketing research, and business data analysis, three disciplines that quietly anchor the behavioural side of this PhD proposal. I then moved to Aston University on the **Aston Enterprise Scholarship 2023**, a full-tuition competitive award, and completed my MSc in Business Analytics with a **First Class (1:1)** classification. Distinctions in Decision Models (mathematical programming, mixed-integer linear programming, multinomial logit discrete-choice modelling, scheduling), Software Analytics (system design, simulation), and Effective Management Consultancy shaped my quantitative toolkit and primed me for the optimisation-plus-simulation backbone this PhD requires.

Logistics and transportation modelling expertise

Logistics and transportation modelling, in the specific senses of mathematical optimisation (mixed-integer linear programming for vehicle-routing variants), behavioural simulation (agent-based and discrete-event), discrete-choice modelling, Digital Twin architecture, and AI/LLM augmentation of decision rules, is the methodological anchor of this application. The work below is mapped one-to-one against the methods named in the project posting.

Supply-chain modelling under Dr. Al-Bazi (MSc dissertation, 2023-2024). My MSc dissertation, *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach*, supervised by Dr. Ammar Al-Bazi over twelve months, applied DeBERTa-v3 transformer NLP and an Ethereum smart-contract architecture to

asymmetric information across three-tier supplier networks. The project required formalising information flows across logistics tiers, modelling supplier behaviour under transparency mechanisms, and validating findings against the supply-chain visibility and bullwhip-mitigation literature. The dissertation's methodological core, an LLM grounded in domain literature combined with a behavioural decision layer, is the direct precursor of the LLM-augmented shipper-decision-rule component proposed in my Research Statement.

Vehicle Routing with Occasional Drivers (VRPOD): MILP formulation. I have implemented the foundational VRPOD model from Archetti, Savelsbergh & Speranza (2016) in `scipy.optimize.milp` with the HiGHS branch-and-bound solver: binary truck-routing and crowd-shipper-assignment variables, capacity and detour-budget constraints, and pre-computed Euclidean cost matrices. On a Birmingham-inspired toy instance (8 customers, 2 trucks, 3 crowd-shippers), enabling occasional-driver assignment **reduces total truck cost by 38.5%**, replicating the savings band reported by Archetti et al. and by Sampaio et al.'s (2019) survey of crowdsourced freight formulations. The implementation, route plots, and detour heatmaps are at github.com/nauvalZulfikar/VRP-MILP.

Crowd-shipping agent-based simulation with behavioural parameter sweep. Using the mesa framework, I built an agent-based model of crowd-shipper acceptance behaviour parameterised on the willingness-to-detour and reward-elasticity estimates of Punel & Stathopoulos (2017). A systematic 450-run sweep across nine reward levels (£10 to £50) and five supply densities ($n = 20$ to 100 commuters) characterises the response surface and locates the reward-saturation inflection in the £25 to £30 range, consistent with the stated-preference findings of Punel & Stathopoulos in their Chicago and Singapore samples. Acceptance ranges from 45.3% ($n = 20$) to 79.8% ($n = 100$): a 34-percentage-point supply-density gradient that quantifies the urban-density effect highlighted by Le, Stathopoulos, Van Woensel & Ukkusuri (2019). Repository: github.com/nauvalZulfikar/Crowd-Shipping-ABM.

Mini Digital Twin for Integrated Passenger-Freight Transport. A three-layer Digital Twin prototype implementing the Grieves & Vickers (2017) physical-virtual-data architecture: a FastAPI mock-IoT layer publishing simulated GPS, demand, and weather telemetry over WebSocket; a coupled mesa agent-based model (passenger and shipper agents) and simpy discrete-event simulation engine (terminal operations); and a Streamlit operator dashboard with live folium map, plotly KPIs, and an interactive disruption-injection lab (rain, vehicle breakdown, demand surge). Smoke-tested: 45% acceptance under base parameters, 62% terminal utilisation, six disruption types each producing behaviourally-responsive cascades. The architecture maps almost line-for-line onto the IPFT Digital Twin description in the project posting. Repository: github.com/nauvalZulfikar/Mini-Digital-Twin.

LLM-grounded shipper-decision-rule generator. An end-to-end system that takes a traveller profile (commute distance, mode, time budget, baseline trust) and emits a literature-grounded crowd-shipper decision rule, using OpenAI gpt-4o-mini augmented with retrieval over a curated behavioural knowledge base of Punel & Stathopoulos (2017), Archetti et al. (2016), and Le et al. (2019). The system falls back to a fully traceable rule-based decision engine when no LLM key is available, a useful property for reproducible academic experimentation. Repository: github.com/nauvalZulfikar/LLM-Shipper-Profiles.

Operations-research-adjacent professional work. As Data Scientist and Project Manager at the Bandung Public Works and Spatial Planning Department (PUTR) since September 2022, I have led ETL and dashboard automation across spatial-planning, building-permits, and tourism-flow datasets, working with the same origin-destination and temporal-flow data shapes that underlie urban transport demand modelling, reducing project delivery time by 45%. I designed and shipped a production-scheduling system with Gantt visualisation and real-time utilisation analytics that lifted planning efficiency by 63%, a discrete-resource scheduling problem from the same combinatorial family as the vehicle-routing problems central to this PhD. I produced a machine learning feasibility study for an automated dam-and-irrigation-gate control system in Cibeureum, framed as a proxy Digital Twin for hydraulic infrastructure under exogenous (rainfall, downstream demand) shocks. Earlier industry work in licensed banking at Bank Muamalat Indonesia (NLP for credit-risk and portfolio analysis under formal model-governance practice), in a UK start-up at Syncwell (predictive ML, A/B testing, +40% CTR), and in a UK non-profit at PCOS Challenge (customer attribution, document automation, -80% manual workload) gave me five years of applied ML rigour inside regulated environments: transferable preparation for the model-governance and validation requirements of urban-transport research.

Why Aston and these supervisors

Aston Business School's positioning of operations research, behavioural management research, and applied simulation under a single departmental roof is rare among UK business schools. Dr. Sajadi's recent integration of Lean Six Sigma with hybrid discrete-event and agent-based simulation in healthcare process optimisation models exactly the methodological combination this proposal requires. Dr. Al-Bazi's recent work on agent-based heuristics for manufacturing scheduling demonstrates fluency in the precise seam, agent-based simulation coupled with classical optimisation, at which the proposed PhD sits. As a returning Aston alumnus with a 1:1 MSc, the Enterprise Scholarship, and an MSc dissertation already supervised by Dr. Al-Bazi on the LLM-supply-chain methodology now central to this proposal, I bring institutional fluency, methodological continuity, and a documented track record that few applicants can match.

What I bring, and what I am still building

I bring **logistics and transportation modelling artefacts** (a VRPOD MILP, a crowd-shipping ABM, an IPFT Digital Twin, an LLM shipper-decision generator) that map one-to-one against the methods named in the project posting; **applied rigour from regulated industry** (banking, government, UK start-up, NPO); and **honest interdisciplinarity** from living and working across Indonesia, Japan, and the UK. I am clear-eyed about the present gap: formal industrial-tool fluency in AnyLogic is not yet a demonstrated strength. I have begun closing this with AnyLogic Personal Learning Edition tutorials and a three-month structured ramp-up plan documented in my Research Statement. I would rather flag this honestly now than discover it in Year One.

Career vision

After the PhD I want to bring Digital-Twin-grade logistics analytics to the organisations that actually move people and parcels: Transport for West Midlands, Transport for London, an Aston-affiliated research centre, or an analytics function at a UK logistics operator. Beyond that, I want to be the researcher who carries this thinking back to the Global South. Bandung's traffic does not have a Digital Twin. By the end of this PhD, I hope to be one of the few researchers in a position to argue that it should, and to build the framework that proves it.